

# xD-69 Hardware Description



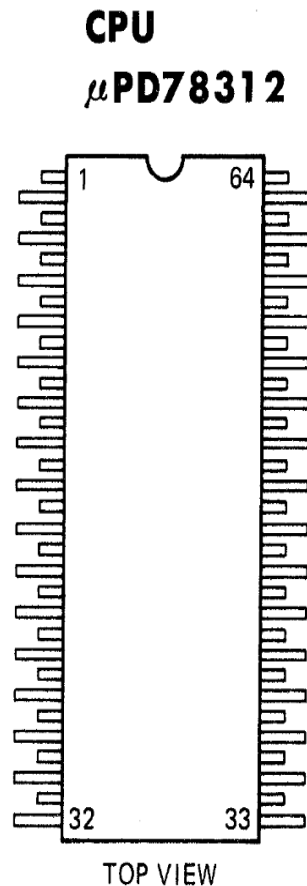
This amazing logo was designed by someone called kilésengati on Discord :)

# CPU: NEC $\mu$ PD78312 – 8/16 bit embedded microcontroller

NEC 78k/3 series microcontroller for embedded devices. CISC(Complex instruction set computer) processor, little endian.

No current ROM dump. Has a PROM that contains opcode stuff I think

Serial MIDI transmission uses RXD and TXD CPU pins.



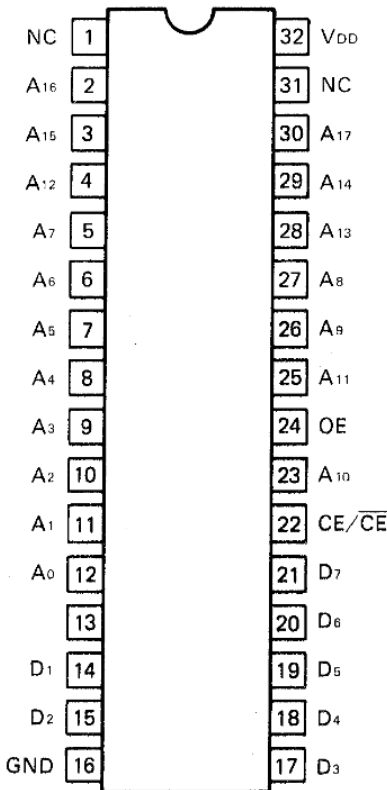
| Pin No. | Pin Name | Input/Output | Description                  | Pin No. | Pin Name | Input/Output | Description          |
|---------|----------|--------------|------------------------------|---------|----------|--------------|----------------------|
| 1       | P0.0     | O            | Port 0                       | 33      | AN0      | I            | Analog input         |
| 2       | P0.1     | O            | Port 0                       | 34      | AN1      | I            | Analog input         |
| 3       | P0.2     | O            | Port 0                       | 35      | AN2      | I            | Analog input         |
| 4       | P0.3     | O            | Port 0                       | 36      | AN3      | I            | Analog input         |
| 5       | P0.4     | O            | Port 0                       | 37      | AVREF    | O            | Reference voltage    |
| 6       | P0.5     | O            | Port 0                       | 38      | AVSS     | -            | Analog VSS           |
| 7       | P0.6     | O            | Port 0                       | 39      | P3.4     | I/O(NC)      | Port 3               |
| 8       | P0.7     | O            | Port 0                       | 40      | P3.5     | I/O(NC)      | Port 3               |
| 9       | P1.0     | I/O          | Port 1                       | 41      | P3.6     | I/O          | Port 3               |
| 10      | P1.1     | I/O          | Port 1                       | 42      | P3.7     | I/O          | Port 3               |
| 11      | P1.2     | I/O          | Port 1                       | 43      | A8       | O            | Address bus          |
| 12      | P1.3     | I/O          | Port 1                       | 44      | A9       | O            | Address bus          |
| 13      | P1.4     | I/O          | Port 1                       | 45      | A10      | O            | Address bus          |
| 14      | P1.5     | I/O          | Port 1                       | 46      | A11      | O            | Address bus          |
| 15      | P1.6     | I/O          | Port 1                       | 47      | A12      | O            | Address bus          |
| 16      | P1.7     | I/O(NC)      | Port 1                       | 48      | A13      | O            | Address bus          |
| 17      | NMI      | I            | Non-maskable interrupt       | 49      | A14      | O            | Address bus          |
| 18      | INT0     | I            | Interrupt from externals     | 50      | A15      | O            | Address bus          |
| 19      | INT1     | I            | Interrupt from externals     | 51      | EA       | I            | External access      |
| 20      | INT2     | I            | Interrupt from externals     | 52      | RESET    | I            | Reset                |
| 21      | TXD      | O            | Transfer serial data         | 53      | RD       | O            | Read strobe          |
| 22      | RXD      | I            | Receive serial data          | 54      | WR       | O            | Write strobe         |
| 23      | SCK      | O(NC)        | Serial transmit clock output | 55      | ALE      | O            | Address latch enable |
| 24      | CTS      | I/O          | Clear to send                | 56      | AD0      | I/O          | Address/data bus     |
| 25      | RFSH     | O(NC)        | Refresh                      | 57      | AD1      | I/O          | Address/data bus     |
| 26      | P3.0     | I            | Port 3                       | 58      | AD2      | I/O          | Address/data bus     |
| 27      | P3.1     | I            | Port 3                       | 59      | AD3      | I/O          | Address/data bus     |
| 28      | P3.2     | I            | Port 3                       | 60      | AD4      | I/O          | Address/data bus     |
| 29      | P3.3     | I            | Port 3                       | 61      | AD5      | I/O          | Address/data bus     |
| 30      | X1       | I            | Crystal                      | 62      | AD6      | I/O          | Address/data bus     |
| 31      | X2       | I            | Crystal                      | 63      | AD7      | I/O          | Address/data bus     |
| 32      | VSS      | -            | Ground                       | 64      | VDD      | -            |                      |

PCM storage ROM – contains PCM partials for the preset playback. Has been ROM dumped. Roland owns the samples so you need to provide your own copy

# Read-Only Memories

OS ROM - This is where the D-50 runs its code from for the D-50 operating system. I have provided the disassembly in another folder. I managed to disassemble the binary using MAME disasm command line

## PCM ROM A/B TC532000

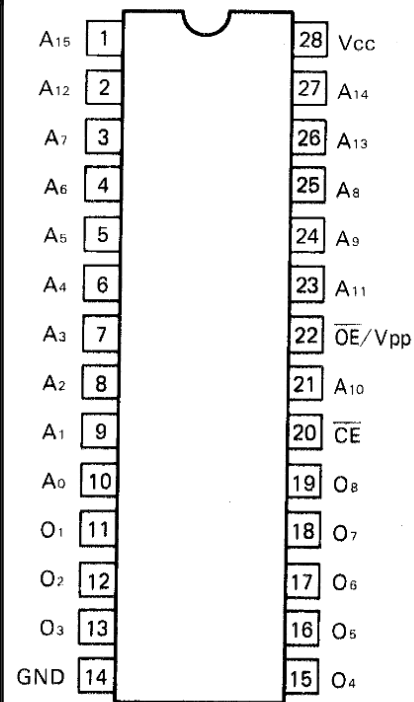


TOP VIEW

| TC532000 ROM(PCM) |          |              |                     |
|-------------------|----------|--------------|---------------------|
| Pin No.           | Pin Name | Input/Output | Description         |
| 1                 | NC       | -            | No connection       |
| 2                 | A16      | I            | Address inputs      |
| 3                 | A15      | I            | Address inputs      |
| 4                 | A12      | I            | Address inputs      |
| 5                 | A7       | I            | Address inputs      |
| 6                 | A6       | I            | Address inputs      |
| 7                 | A5       | I            | Address inputs      |
| 8                 | A4       | I            | Address inputs      |
| 9                 | A3       | I            | Address inputs      |
| 10                | A2       | I            | Address inputs      |
| 11                | A1       | I            | Address inputs      |
| 12                | A0       | I            | Address inputs      |
| 13                | D0       | O            | Data outputs        |
| 14                | D1       | O            | Data outputs        |
| 15                | D2       | O            | Data outputs        |
| 16                | GND      | -            | Ground              |
| 17                | D3       | O            | Data outputs        |
| 18                | D4       | O            | Data outputs        |
| 19                | D5       | O            | Data outputs        |
| 20                | D6       | O            | Data outputs        |
| 21                | D7       | O            | Data outputs        |
| 22                | CE/CE    | I            | Chip enable input   |
| 23                | A10      | I            | Address inputs      |
| 24                | OE       | I            | Output enable input |
| 25                | A11      | I            | Address inputs      |
| 26                | A9       | I            | Address inputs      |
| 27                | A8       | I            | Address inputs      |
| 28                | A13      | I            | Address inputs      |
| 29                | A14      | I            | Address inputs      |
| 30                | A17      | I            | Address inputs      |
| 31                | NC       | -            | No connection       |
| 32                | VDD      | -            | Power               |

| MBM27C512 ROM (OS) |          |              |                     |
|--------------------|----------|--------------|---------------------|
| Pin No.            | Pin Name | Input/Output | Description         |
| 1                  | A15      | I            | Address input       |
| 2                  | A12      | I            | Address input       |
| 3                  | A7       | I            | Address input       |
| 4                  | A6       | I            | Address input       |
| 5                  | A5       | I            | Address input       |
| 6                  | A4       | I            | Address input       |
| 7                  | A3       | I            | Address input       |
| 8                  | A2       | I            | Address input       |
| 9                  | A1       | I            | Address input       |
| 10                 | A0       | I            | Address input       |
| 11                 | O1       | O            | Data output         |
| 12                 | O2       | O            | Data output         |
| 13                 | O3       | O            | Data output         |
| 14                 | GND      | -            | Ground              |
| 15                 | O4       | O            | Data output         |
| 16                 | O5       | O            | Data output         |
| 17                 | O6       | O            | Data output         |
| 18                 | O7       | O            | Data output         |
| 19                 | O8       | O            | Data output         |
| 20                 | CE       | I            | Chip enable input   |
| 21                 | A10      | I            | Address input       |
| 22                 | OE/VPP   | I            | Output enable input |
| 23                 | A11      | I            | Address input       |
| 24                 | A9       | I            | Address input       |
| 25                 | A8       | I            | Address input       |
| 26                 | A13      | I            | Address input       |
| 27                 | A14      | I            | Address input       |
| 28                 | VCC      | -            | Power               |

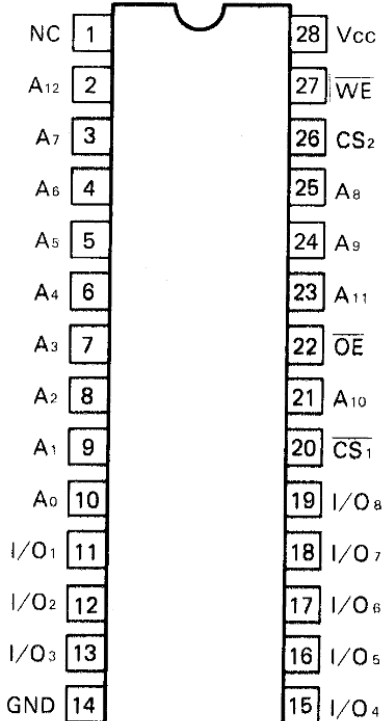
## EP ROM MBM27C512



TOP VIEW

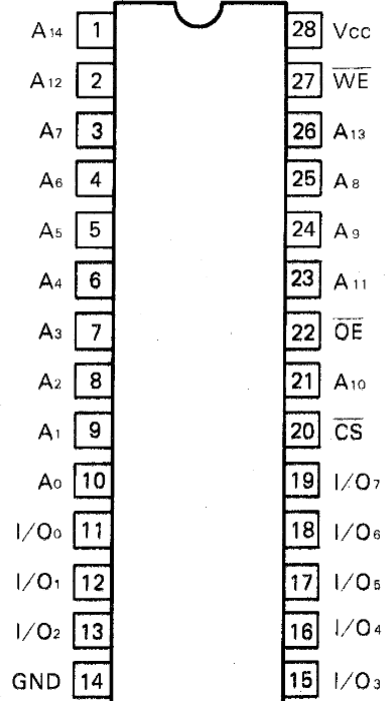
# Random-Access Memories

**S RAM  
HM6264ASP**



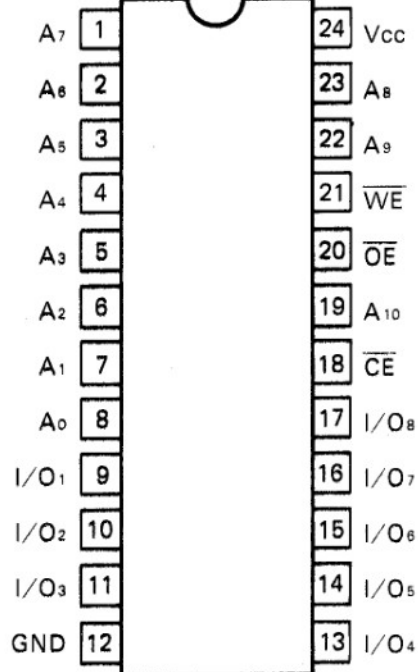
TOP VIEW

**S RAM  
HM62256LP**



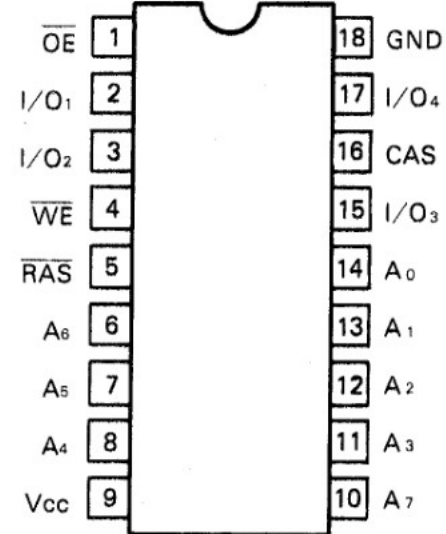
TOP VIEW

**S RAM  
LC3517AS**



TOP VIEW

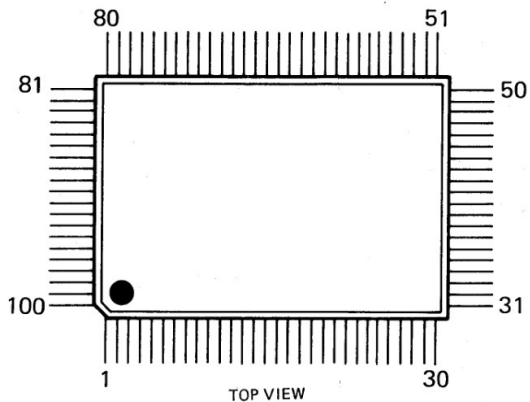
**D RAM  
 $\mu$ PD41416**



TOP VIEW

# Target Custom ASIC – Chorus DSP

**CHORUS CUSTOM IC  
MB87137**

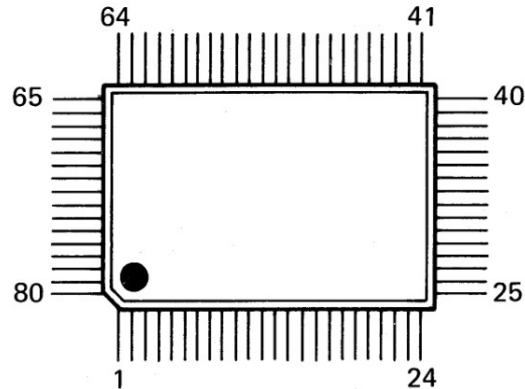


Please see:  
<https://github.com/pxlsr/FujiCells>  
for a detailed and comprehensive  
documentation of the ASICs

| Pin No. | Pin Name | Input/Output | Description                  | Pin No. | Pin Name | Input/Output | Description                                |
|---------|----------|--------------|------------------------------|---------|----------|--------------|--------------------------------------------|
| 1       | RES      | I            | Reset input                  | 51      | O8       | O            | Data input                                 |
| 2       | E        | I            | Chip enable input            | 52      | O9       | O            | Data input                                 |
| 3       | VDD      | -            | +5v                          | 53      | VDD      | -            | +5v                                        |
| 4       | CS       | I            | Chip Select input            | 54      | O10      | O            | Data input                                 |
| 5       | RW       | I            | Write Pulse input            | 55      | O11      | O            | Data input                                 |
| 6       | RD       | I            | Read Pulse input             | 56      | O12      | O            | Data input                                 |
| 7       | CS       | I            | Chip select input            | 57      | O13      | O            | Data input                                 |
| 8       | A0       | I            | Connect to CPU address bus   | 58      | O14      | O            | Data input                                 |
| 9       | A1       | I            | Connect to CPU address bus   | 59      | O15      | O            | Data input                                 |
| 10      | A2       | I            | Connect to CPU address bus   | 60      | RA0      | O            | Connects to SRAM address bus RA13 not used |
| 11      | D0       | I/O          | Connect to CPU data bus      | 61      | WE       | O            | SRAM write pulse output                    |
| 12      | D1       | I/O          | Connect to CPU data bus      | 62      | RA1      | O            | Connects to SRAM address bus RA13 not used |
| 13      | D2       | I/O          | Connect to CPU data bus      | 63      | RA2      | O            | Connects to SRAM address bus RA13 not used |
| 14      | D3       | I/O          | Connect to CPU data bus      | 64      | RA3      | O            | Connects to SRAM address bus RA13 not used |
| 15      | VSS      | -            | GND                          | 65      | VSS      | -            | GND                                        |
| 16      | D4       | I/O          | Connect to CPU data bus      | 66      | RA4      | O            | Connects to SRAM address bus RA13 not used |
| 17      | D5       | I/O          | Connect to CPU data bus      | 67      | RA5      | O            | Connects to SRAM address bus RA13 not used |
| 18      | D6       | I/O          | Connect to CPU data bus      | 68      | RA6      | O            | Connects to SRAM address bus RA13 not used |
| 19      | D7       | I/O          | Connect to CPU data bus      | 69      | RA7      | O            | Connects to SRAM address bus RA13 not used |
| 20      | DOE      | I            | Data out enable input        | 70      | RA8      | O            | Connects to SRAM address bus RA13 not used |
| 21      | INCK     | I            | Input data latch clock input | 71      | OE       | O            | SRAM out enable output                     |
| 22      | SIN      | I            | Sync input                   | 72      | RA9      | O            | Connects to SRAM address bus RA13 not used |
| 23      | SOUT     | O            | Sync output                  | 73      | RA10     | O            | Connects to SRAM address bus RA13 not used |
| 24      | LRS      | I            | L/R select input             | 74      | RA11     | O            | Connects to SRAM address bus RA13 not used |
| 25      | I0       | I            | Data input                   | 75      | CE       | O            | SRAM chip enable output                    |
| 26      | I1       | I            | Data input                   | 76      | RA12     | O            | Connects to SRAM address bus RA13 not used |
| 27      | I2       | I            | Data input                   | 77      | PD7      | I/O          | Connect to SRAM data bus                   |
| 28      | VDD      | -            | +5v                          | 78      | VDD      | -            | +5v                                        |
| 29      | I3       | I            | Data input                   | 79      | RA13     | O            | Connects to SRAM address bus RA13 not used |
| 30      | I4       | I            | Data input                   | 80      | PD6      | I/O          | Connect to SRAM data bus                   |
| 31      | I5       | I            | Data input                   | 81      | PD5      | I/O          | Connect to SRAM data bus                   |
| 32      | I6       | I            | Data input                   | 82      | PD4      | I/O          | Connect to SRAM data bus                   |
| 33      | I7       | I            | Data input                   | 83      | PD3      | I/O          | Connect to SRAM data bus                   |
| 34      | I8       | I            | Data input                   | 84      | PD2      | I/O          | Connect to SRAM data bus                   |
| 35      | I9       | I            | Data input                   | 85      | PD1      | I/O          | Connect to SRAM data bus                   |
| 36      | I10      | I            | Data input                   | 86      | PD0      | I/O          | Connect to SRAM data bus                   |
| 37      | I11      | I            | Data input                   | 87      | VSS      | -            | GND                                        |
| 38      | I12      | I            | Data input                   | 88      | X1       | I            | Master clock input                         |
| 39      | I13      | I            | Data input                   | 89      | X2       | O            | not used                                   |
| 40      | VSS      | -            | GND                          | 90      | VSS      | -            | GND                                        |
| 41      | I14      | I            | Data input                   | 91      | ROMT     | I            | No description                             |
| 42      | I15      | I            | Data input                   | 92      | RAMT     | I            | No description                             |
| 43      | O0       | O            | Data input                   | 93      | CTRT     | I            | No description                             |
| 44      | O1       | O            | Data input                   | 94      | THRU     | I            | No description                             |
| 45      | O2       | O            | Data input                   | 95      | ECTL     | I            | External control select input              |
| 46      | O3       | O            | Data input                   | 96      | ADDA     | I            | No description                             |
| 47      | O4       | O            | Data input                   | 97      | OFST     | I            | offset binary select input                 |
| 48      | O5       | O            | Data input                   | 98      | PSFT     | I            | No description                             |
| 49      | O6       | O            | Data input                   | 99      | LHLD     | O            | Signal output for S/H – not used           |
| 50      | O7       | O            | Data input                   | 100     | RHLD     | O            | Signal output for S/H – not used           |

# Target Custom ASIC – Reverb DSP

## REVERB CUSTOM IC MB87126-006



TOP VIEW

| Pin No. | Pin Name | Input/Output | Description                                                             | Pin No. | Pin Name | Input/Output | Description                                          |
|---------|----------|--------------|-------------------------------------------------------------------------|---------|----------|--------------|------------------------------------------------------|
| 1       | DC0      | O            | Data output for chorus chip and DAC                                     | 51      | DR11     | I/O          | Connect to RAM data bus, Synth and Chorus data input |
| 2       | DC1      | O            | Data output for chorus chip and DAC                                     | 52      | VSS      | -            | GND                                                  |
| 3       | STR1     | I            | Pulled low                                                              | 53      | DR12     | I/O          | Connect to RAM data bus, Synth and Chorus data input |
| 4       | DIN      | I            | Pulled low                                                              | 54      | DR13     | I/O          | Connect to RAM data bus, Synth and Chorus data input |
| 5       | CLEA     | I            | Pulled low                                                              | 55      | DR14     | I/O          | Connect to RAM data bus, Synth and Chorus data input |
| 6       | RD0      | O            | Control output for enable and for S/H and Lower four bit D/A conversion | 56      | DR15     | I/O          | Connect to RAM data bus, Synth and Chorus data input |
| 7       | RD1      | O            | Control output for enable and for S/H and Lower four bit D/A conversion | 57      | DR16     | I/O          | Connect to RAM data bus, Synth and Chorus data input |
| 8       | RD2      | O            | Control output for enable and for S/H and Lower four bit D/A conversion | 58      | DR17     | I/O          | Connect to RAM data bus, Synth and Chorus data input |
| 9       | RD3      | O            | Control output for enable and for S/H and Lower four bit D/A conversion | 59      | DR18     | I/O          | Connect to RAM data bus, Synth and Chorus data input |
| 10      | RD4      | O            | Control output for enable and for S/H and Lower four bit D/A conversion | 60      | DR19     | I/O          | Connect to RAM data bus, Synth and Chorus data input |
| 11      | RSET     | I            | Pulled low                                                              | 61      | DR20     | I/O          | Connect to RAM data bus, Synth and Chorus data input |
| 12      | VSS      | -            | GND                                                                     | 62      | DR21     | I/O          | Connect to RAM data bus, Synth and Chorus data input |
| 13      | SLRQ     | I            | Pulled low                                                              | 63      | DR22     | I/O          | Connect to RAM data bus, Synth and Chorus data input |
| 14      | MSCK     | I            | Master clock input                                                      | 64      | DR23     | I/O          | Connect to RAM data bus, Synth and Chorus data input |
| 15      | VSS      | -            | GND                                                                     | 65      | VSS      | -            | GND                                                  |
| 16      | SLCK     | O            | Not used                                                                | 66      | DC2      | O            | Data output for chorus chip and DAC                  |
| 17      | TEST     | I            | Pulled low                                                              | 67      | DC3      | O            | Data output for chorus chip and DAC                  |
| 18      | TMB      | O            | Time base signal output                                                 | 68      | DC4      | O            | Data output for chorus chip and DAC                  |
| 19      | VDD      | -            | +5V                                                                     | 69      | DC5      | O            | Data output for chorus chip and DAC                  |
| 20      | LOAD     | O            | Sync signal output                                                      | 70      | DC6      | O            | Data output for chorus chip and DAC                  |
| 21      | SYNC     | I            | Sync signal input                                                       | 71      | DC7      | O            | Data output for chorus chip and DAC                  |
| 22      | INCK     | I            | Data latch clock input for initialization                               | 72      | DC8      | O            | Data output for chorus chip and DAC                  |
| 23      | ERCL     | I            | Busy reset output                                                       | 73      | VDD      | -            | +5V                                                  |
| 24      | BUSY     | O            | Serial data transfer error output (parity check)                        | 74      | DC9      | O            | Data output for chorus chip and DAC                  |
| 25      | SXD      | I            | Serial data input                                                       | 75      | DC10     | O            | Data output for chorus chip and DAC                  |
| 26      | SCK      | I            | Serial data read-in clock input                                         | 76      | DC11     | O            | Data output for chorus chip and DAC                  |
| 27      | DA0      | O            | Connect to RAM address bus                                              | 77      | DC12     | O            | Data output for chorus chip and DAC                  |
| 28      | DA1      | O            | Connect to RAM address bus                                              | 78      | DC13     | O            | Data output for chorus chip and DAC                  |
| 29      | DA2      | O            | Connect to RAM address bus                                              | 79      | DC14     | O            | Data output for chorus chip and DAC                  |
| 30      | DA3      | O            | Connect to RAM address bus                                              | 80      | DC15     | O            | Data output for chorus chip and DAC                  |
| 31      | DA4      | O            | Connect to RAM address bus                                              |         |          |              |                                                      |
| 32      | DA5      | O            | Connect to RAM address bus                                              |         |          |              |                                                      |
| 33      | VDD      | -            | +5V                                                                     |         |          |              |                                                      |
| 34      | DA6      | O            | Connect to RAM address bus                                              |         |          |              |                                                      |
| 35      | DA7      | O            | Connect to RAM address bus                                              |         |          |              |                                                      |
| 36      | VSS      | -            | GND                                                                     |         |          |              |                                                      |
| 37      | RAS      | O            | Row address strobe output                                               |         |          |              |                                                      |
| 38      | WE       | O            | DRAM write pulse output                                                 |         |          |              |                                                      |
| 39      | CAS      | O            | Column address strobe output                                            |         |          |              |                                                      |
| 40      | DR0      | I/O          | Connect to RAM data bus, Synth and Chorus data input                    |         |          |              |                                                      |
| 41      | DR1      | I/O          | Connect to RAM data bus, Synth and Chorus data input                    |         |          |              |                                                      |
| 42      | DR2      | I/O          | Connect to RAM data bus, Synth and Chorus data input                    |         |          |              |                                                      |
| 43      | DR3      | I/O          | Connect to RAM data bus, Synth and Chorus data input                    |         |          |              |                                                      |
| 44      | DR4      | I/O          | Connect to RAM data bus, Synth and Chorus data input                    |         |          |              |                                                      |
| 45      | DR5      | I/O          | Connect to RAM data bus, Synth and Chorus data input                    |         |          |              |                                                      |
| 46      | DR6      | I/O          | Connect to RAM data bus, Synth and Chorus data input                    |         |          |              |                                                      |
| 47      | DR7      | I/O          | Connect to RAM data bus, Synth and Chorus data input                    |         |          |              |                                                      |
| 48      | DR8      | I/O          | Connect to RAM data bus, Synth and Chorus data input                    |         |          |              |                                                      |
| 49      | DR9      | I/O          | Connect to RAM data bus, Synth and Chorus data input                    |         |          |              |                                                      |
| 50      | DR10     | I/O          | Connect to RAM data bus, Synth and Chorus data input                    |         |          |              |                                                      |

Please see:  
<https://github.com/pxlsr/FujiCells>  
 for a detailed and comprehensive  
 documentation of the ASICs

# Target Custom ASIC – Synth (DSP core)

**SYNTH CUSTOM IC  
MB87136**

|    |    |           |    |    |    |    |    |    |    |    |    |
|----|----|-----------|----|----|----|----|----|----|----|----|----|
| 34 | 33 | 32        | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 |
| 35 | 72 | 71        | 70 | 69 | 68 | 67 | 66 | 65 | 64 | 63 | 22 |
| 36 | 73 |           |    |    | 86 | 85 |    |    |    | 62 | 21 |
| 37 | 74 |           |    |    |    |    |    |    |    | 61 | 20 |
| 38 | 75 |           |    |    |    |    |    |    |    | 60 | 19 |
| 39 | 76 | 87        |    |    |    |    |    |    | 84 | 59 | 18 |
| 40 | 77 | 88        |    |    |    |    |    |    | 83 | 58 | 17 |
| 41 | 78 |           |    |    |    |    |    |    |    | 57 | 16 |
| 42 | 79 | INDEX PIN |    |    |    |    |    |    |    | 56 | 15 |
| 43 | 80 |           |    |    | 81 | 82 |    |    |    | 55 | 14 |
| 44 | 45 | 46        | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 13 |
| 1  | 2  | 3         | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 |

TOP VIEW

| Pin No. | Pin Name | Input/Output | Description                        |
|---------|----------|--------------|------------------------------------|
| 1       | CS       | I            | Chip select                        |
| 2       | A0       | I            | Connect to CPU address bus         |
| 3       | A1       | I            | Connect to CPU address bus         |
| 4       | A2       | I            | Connect to CPU address bus         |
| 5       | A3       | I            | Connect to CPU address bus         |
| 6       | A4       | I            | Connect to CPU address bus         |
| 7       | D0       | I/O          | Connect to CPU data bus            |
| 8       | D1       | I/O          | Connect to CPU data bus            |
| 9       | D2       | I/O          | Connect to CPU data bus            |
| 10      | D3       | I/O          | Connect to CPU data bus            |
| 11      | PD0      | I            | Connect to ROM data bus            |
| 12      | PD1      | I            | Connect to ROM data bus            |
| 13      | PD2      | I            | Connect to ROM data bus            |
| 14      | PD3      | I            | Connect to ROM data bus            |
| 15      | RA0      | O            | Connect to ROM address bus         |
| 16      | RA1      | O            | Connect to ROM address bus         |
| 17      | RA2      | O            | Connect to ROM address bus         |
| 18      | RA3      | O            | Connect to ROM address bus         |
| 19      | RA4      | O            | Connect to ROM address bus         |
| 20      | RA5      | O            | Connect to ROM address bus         |
| 21      | RA6      | O            | Connect to ROM address bus         |
| 22      | RA7      | O            | Connect to ROM address bus         |
| 23      | RA8      | O            | Connect to ROM address bus         |
| 24      | RA9      | O            | Connect to ROM address bus         |
| 25      | RA10     | O            | Connect to ROM address bus         |
| 26      | RA11     | O            | Connect to ROM address bus         |
| 27      | O0       | O            | Data output                        |
| 28      | O1       | O            | Data output                        |
| 29      | O2       | O            | Data output                        |
| 30      | O3       | O            | Data output                        |
| 31      | O4       | O            | Data output                        |
| 32      | O5       | O            | Data output                        |
| 33      | O6       | O            | Data output                        |
| 34      | O7       | O            | Data output                        |
| 35      | O8       | O            | Data output                        |
| 36      | SH0      | O            | Not used                           |
| 37      | SH1      | O            | Not used                           |
| 38      | -        | -            | Not used                           |
| 39      | X1       | I/O          | Crystal input (32.768mhz)          |
| 40      | 32M      | O            | The same frequency as master clock |
| 41      | -        | -            | -                                  |
| 42      | SYI      | I            | Sync signal input                  |
| 43      | WR       | I            | Write pulse input                  |
| 44      | INT      | O            | Interrupt output                   |
| 45      | OE       | I            | Output enable input                |

| Pin No. | Pin Name | Input/Output | Description                                                                       |
|---------|----------|--------------|-----------------------------------------------------------------------------------|
| 46      | A5       | I            | Connect to CPU address bus                                                        |
| 47      | A6       | I            | Connect to CPU address bus                                                        |
| 48      | A7       | I            | Connect to CPU address bus                                                        |
| 49      | A8       | I            | Connect to CPU address bus                                                        |
| 50      | D4       | I/O          | Connect to CPU data bus                                                           |
| 51      | D5       | I/O          | Connect to CPU data bus                                                           |
| 52      | D6       | I/O          | Connect to CPU data bus                                                           |
| 53      | D7       | I/O          | Connect to CPU data bus                                                           |
| 54      | PD4      | I            | Connect to ROM data bus                                                           |
| 55      | PD5      | I            | Connect to ROM data bus                                                           |
| 56      | PD6      | I            | Connect to ROM data bus                                                           |
| 57      | PD7      | I            | Connect to ROM data bus                                                           |
| 58      | RA12     | O            | Connect to ROM address bus                                                        |
| 59      | RA13     | O            | Connect to ROM address bus                                                        |
| 60      | RA14     | O            | Connect to ROM address bus                                                        |
| 61      | RA15     | O            | Connect to ROM address bus                                                        |
| 62      | RA16     | O            | Connect to ROM address bus                                                        |
| 63      | RA17     | O            | Connect to ROM address bus                                                        |
| 64      | RA18     | O            | Connect to ROM address bus                                                        |
| 65      | RA19     | O            | Connect to ROM address bus                                                        |
| 66      | O9       | O            | Data output                                                                       |
| 67      | O10      | O            | Data output                                                                       |
| 68      | O11      | O            | Data output                                                                       |
| 69      | O12      | O            | Data output                                                                       |
| 70      | O13      | O            | Data output                                                                       |
| 71      | O14      | O            | Data output                                                                       |
| 72      | O15      | O            | Data output                                                                       |
| 73      | SH2      | O            | Not used                                                                          |
| 74      | SH3      | O            | Not used                                                                          |
| 75      | -        | -            | Not used                                                                          |
| 76      | X2       | I/O          | Crystal input                                                                     |
| 77      | 16M      | O            | Output frequency is half of master clock                                          |
| 78      | CKIN     | I            | Input frequency is a combination of the master clock and one half of master clock |
| 79      | -        | -            | Not used                                                                          |
| 80      | RD       | I            | Read pulse input                                                                  |
| 81      | VSS      | -            | GND                                                                               |
| 82      | VDD      | -            | +5V                                                                               |
| 83      | VDD      | -            | +5V                                                                               |
| 84      | VSS      | -            | GND                                                                               |
| 85      | VSS      | -            | GND                                                                               |
| 86      | VDD      | -            | +5V                                                                               |
| 87      | VDD      | -            | +5V                                                                               |
| 88      | VSS      | -            | GND                                                                               |

Please see:

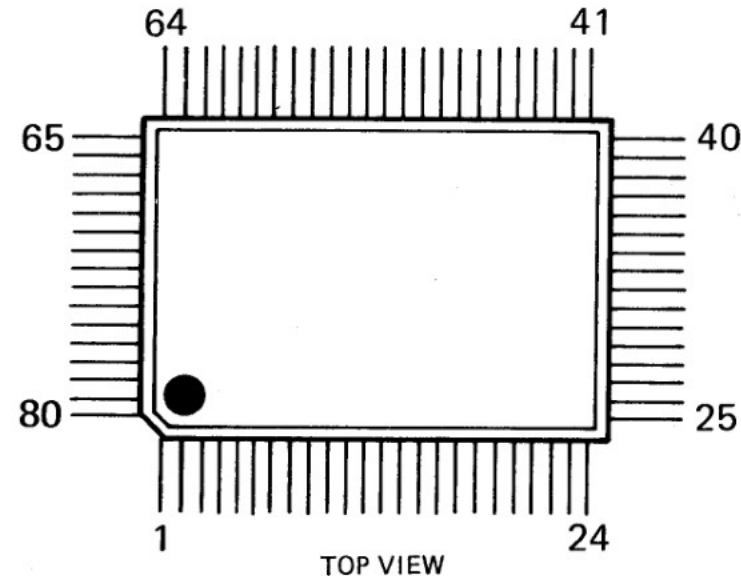
<https://github.com/pxlsr/FujiCells>

for a detailed and comprehensive documentation of the ASICs



# Target Custom Gate Array – LCD controller with possible peripheral control logic?

**GATE ARRAY  
HG61H25B18F**



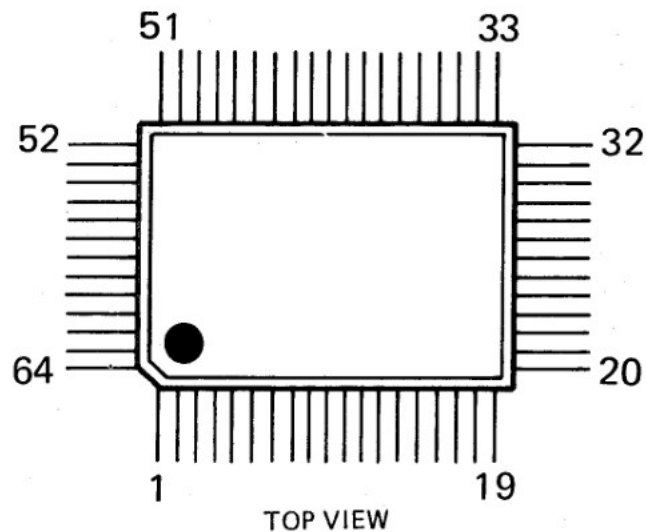
| PIN NO. | NAME  | I/O    | PIN NO. | NAME         | I/O      | PIN NO. | NAME | I/O | PIN NO. | NAME  | I/O     |
|---------|-------|--------|---------|--------------|----------|---------|------|-----|---------|-------|---------|
| 1       | SYNT2 | O (NC) | 21      | ALE          | I        | 41      | EC   | O   | 61      | R2    | I       |
| 2       | IRAM  | O (NC) | 22      | <u>WR</u>    | I        | 42      | O0   | O   | 62      | R3    | I       |
| 3       | RAM   | O      | 23      | <u>RD</u>    | I        | 43      | O1   | O   | 63      | R4    | I       |
| 4       | A7    | O      | 24      | <u>RESET</u> | I        | 44      | O2   | O   | 64      | R5    | I       |
| 5       | A6    | O      | 25      | A15          | I        | 45      | O3   | O   | 65      | R6    | I       |
| 6       | A5    | O      | 26      | A14          | I        | 46      | O4   | O   | 66      | R7    | I       |
| 7       | A4    | O      | 27      | A13          | I        | 47      | O5   | O   | 67      | CORUS | O       |
| 8       | A3    | O      | 28      | A12          | I        | 48      | O6   | O   | 68      | SCK   | O       |
| 9       | A2    | O      | 29      | A11          | I        | 49      | O7   | O   | 69      | SXD   | O       |
| 10      | A1    | O      | 30      | A10          | I        | 50      | S0   | O   | 70      | BUSY  | I       |
| 11      | A0    | O      | 31      | A9           | I        | 51      | S1   | O   | 71      | ERCL  | O       |
| 12      | Vss   | -      | 32      | A8           | I        | 52      | Vss  | -   | 72      | LOAD  | I       |
| 13      | AD7   | I/O    | 33      | VDD          | -        | 53      | S2   | O   | 73      | VDD   | -       |
| 14      | AD6   | I/O    | 34      | ARS          | I (HIGH) | 54      | S3   | O   | 74      | TMB   | I       |
| 15      | AD5   | I/O    | 35      | INT1         | O (NC)   | 55      | S4   | O   | 75      | SINT1 | I (LOW) |
| 16      | AD4   | I/O    | 36      | INT2         | O        | 56      | S5   | O   | 76      | SINT2 | I (LOW) |
| 17      | AD3   | I/O    | 37      | DSCAN        | O        | 57      | S6   | O   | 77      | TEST1 | I (LOW) |
| 18      | AD2   | I/O    | 38      | <u>ERAM</u>  | O        | 58      | S7   | O   | 78      | CLK   | I       |
| 19      | AD1   | I/O    | 39      | ERAM         | O (NC)   | 59      | R0   | I   | 79      | TEST2 | I (LOW) |
| 20      | AD0   | I/O    | 40      | RS           | O        | 60      | R1   | I   | 80      | SYNT1 | O       |



# Target Custom Gate Array – Memory card controller

## GATE ARRAY

**$\mu$ PD65005G-062**

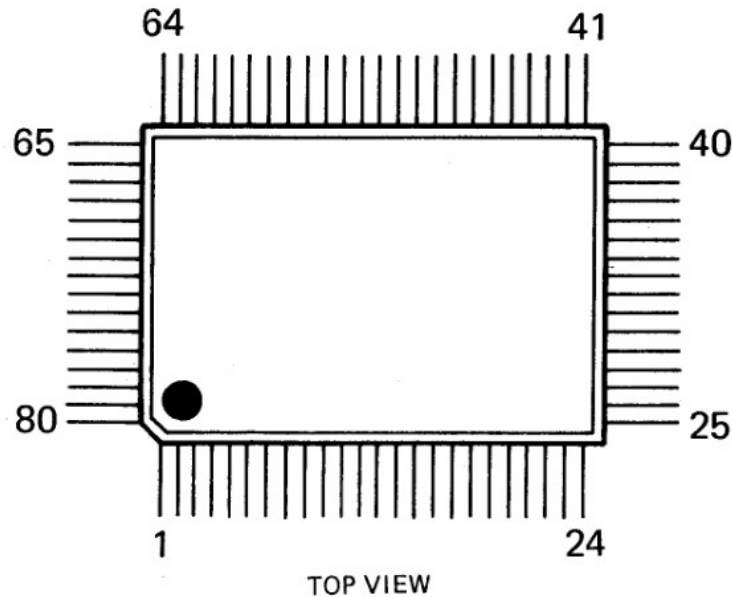


| PIN NO. | NAME                   | I/O | PIN NO. | NAME | I/O     | PIN NO. | NAME | I/O | PIN NO. | NAME                   | I/O     |
|---------|------------------------|-----|---------|------|---------|---------|------|-----|---------|------------------------|---------|
| 1       | NC                     | -   | 17      | NC   | -       | 33      | NC   | -   | 49      | NC                     | -       |
| 2       | NC                     | -   | 18      | NC   | -       | 34      | NC   | -   | 50      | CD0                    | I/O     |
| 3       | AD7                    | I/O | 19      | A13  | I       | 35      | CA5  | O   | 51      | CD1                    | I/O     |
| 4       | AD6                    | I/O | 20      | A12  | I       | 36      | CA6  | O   | 52      | CD2                    | I/O     |
| 5       | AD5                    | I/O | 21      | A11  | I       | 37      | CA7  | O   | 53      | CD3                    | I/O     |
| 6       | AD4                    | I/O | 22      | A10  | I       | 38      | CA8  | O   | 54      | CD4                    | I/O     |
| 7       | AD3                    | I/O | 23      | A9   | I       | 39      | CA9  | O   | 55      | CD5                    | I/O     |
| 8       | AD2                    | I/O | 24      | A8   | I       | 40      | CA10 | O   | 56      | CD6                    | I/O     |
| 9       | AD1                    | I/O | 25      | SEL  | I (LOW) | 41      | CA11 | O   | 57      | CD7                    | I/O     |
| 10      | AD0                    | I/O | 26      | Vss  | -       | 42      | CA12 | O   | 58      | Vss                    | -       |
| 11      | Vss                    | -   | 27      | VDD  | -       | 43      | CA13 | O   | 59      | VDD                    | -       |
| 12      | VDD                    | -   | 28      | CA0  | O       | 44      | CA14 | O   | 60      | BATT                   | I (LOW) |
| 13      | ALE                    | I   | 29      | CA1  | O       | 45      | MR   | O   | 61      | SENS                   | I (NC)  |
| 14      | $\overline{\text{WR}}$ | I   | 30      | CA2  | O       | 46      | CWR  | O   | 62      | RCS                    | I       |
| 15      | $\overline{\text{RD}}$ | I   | 31      | CA3  | O       | 47      | CCS  | O   | 63      | $\overline{\text{CS}}$ | I       |
| 16      | A14                    | I   | 32      | CA4  | O       | 48      | CRD  | O   | 64      | NC                     | -       |

# Target Custom Gate Array

## – Keyboard controller

**GATE ARRAY  
MB63H149**



| PIN NO. | NAME | I/O | PIN NO. | NAME              | I/O      | PIN NO. | NAME | I/O      | PIN NO. | NAME             | I/O |
|---------|------|-----|---------|-------------------|----------|---------|------|----------|---------|------------------|-----|
| 1       | T7   | O   | 21      | BR9               | I        | 41      | AD7  | I/O      | 61      | RA1              | O   |
| 2       | BR0  | I   | 22      | MK9               | I        | 42      | CA8  | I        | 62      | RA10             | O   |
| 3       | MK0  | I   | 23      | BR10              | I        | 43      | CA9  | I        | 63      | RA2              | O   |
| 4       | BR1  | I   | 24      | MK10              | I        | 44      | CA10 | I (LOW)  | 64      | $\overline{ROE}$ | I/O |
| 5       | MK1  | I   | 25      | RES               | I        | 45      | CS   | I        | 65      | RA3              | O   |
| 6       | BR2  | I   | 26      | $\overline{EXCK}$ | I/O      | 46      | XT1  | I        | 66      | $\overline{RWE}$ | O   |
| 7       | MK2  | I   | 27      | E                 | I (HIGH) | 47      | XT2  | O (NC)   | 67      | RA4              | O   |
| 8       | BR3  | I   | 28      | INT               | O        | 48      | ASEL | O (NC)   | 68      | RA9              | O   |
| 9       | MK3  | I   | 29      | AS                | I        | 49      | MOD1 | I (HIGH) | 69      | RA5              | O   |
| 10      | BR4  | I   | 30      | $\overline{CRES}$ | O (NC)   | 50      | MOD2 | I (LOW)  | 70      | RA8              | O   |
| 11      | MK4  | I   | 31      | $\overline{CRNW}$ | I        | 51      | RD3  | I/O      | 71      | RA6              | O   |
| 12      | Vss  | -   | 32      | SRCK              | O (NC)   | 52      | Vss  | -        | 72      | RA7              | O   |
| 13      | BR5  | I   | 33      | VDD               | -        | 53      | RD4  | I/O      | 73      | VDD              | -   |
| 14      | MK5  | I   | 34      | AD0               | I/O      | 54      | RD2  | I/O      | 74      | T0               | O   |
| 15      | BR6  | I   | 35      | AD1               | I/O      | 55      | RD5  | I/O      | 75      | T1               | O   |
| 16      | MK6  | I   | 36      | AD2               | I/O      | 56      | RD1  | I/O      | 76      | T2               | O   |
| 17      | BR7  | I   | 37      | AD3               | I/O      | 57      | RD6  | I/O      | 77      | T3               | O   |
| 18      | MK7  | I   | 38      | AD4               | I/O      | 58      | RD0  | I/O      | 78      | T4               | O   |
| 19      | BR8  | I   | 39      | AD5               | I/O      | 59      | RD7  | I/O      | 79      | T5               | O   |
| 20      | MK8  | I   | 40      | AD6               | I/O      | 60      | RA0  | O        | 80      | T6               | O   |